

Highlights

Water transparency retrieval in a high-altitude large lake from Sentinel-2 and Landsat 8/9: cross-sensor harmonization limits, validation design, and calibrated uncertainty in Lake Titicaca

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- Land-derived Landsat/Sentinel-2 harmonization is invalid over dark oligotrophic water
- Sensors stay 99.8% separable after harmonization; intercepts exceed signal 70-fold
- Campaign, not station, is the blocking unit in campaign-based monitoring programmes
- Honest Secchi retrieval: $R^2=0.57$, RMSE 1.86 m over 812 real match-ups
- Conformal intervals are calibrated marginally but under-cover in the clearest water

Water transparency retrieval in a high-altitude large lake from Sentinel-2 and Landsat 8/9: cross-sensor harmonization limits, validation design, and calibrated uncertainty in Lake Titicaca

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Abstract

Lake Titicaca (8372 km², 3810 m a.s.l.) is the largest lake in South America and a transboundary freshwater reserve for millions of people, yet its water quality is documented only by infrequent field campaigns. We assembled 1002 satellite–field match-ups (340 Sentinel-2, 662 Landsat 8/9) against twelve years of IMARPE limnological monitoring and used them to ask what a defensible multi-mission Secchi disk depth (Z_{SD}) product looks like in clear oligotrophic high-altitude water. Two widely used practices fail here. First, the cross-sensor harmonization coefficients of Roy et al. (2016), fitted over terrestrial targets and routinely reused to build harmonized Landsat/Sentinel-2 archives, are invalid over dark water: their additive intercepts (0.0116–0.0448 in reflectance) exceed the native over-water Landsat signal by factors of 1.4 to 69.8, and a classifier still identifies the source sensor with 99.8% accuracy after the transform (chance 66.3%). Refitting locally on 270 same-day paired observations restores agreement in green ($r = 0.71$) but not in blue ($r = 0.26$) — the band the retrieval most depends on. Second, blocking cross-validation by monitoring station, the usual remedy for spatial autocorrelation, retains nothing in a campaign-based programme: under campaign-blocked folds a station-mean null still reaches $R^2 = 0.208$ and a zone-mean null $R^2 = 0.287$, while the campaign-mean null collapses to -0.012 . Under campaign-blocked validation a Random Forest retrieves Z_{SD} with $R^2 = 0.574$ and RMSE 1.86 m ($n = 812$), against 0.097 for a correctly bounded classical blue/green band ratio. Split-conformal intervals are well calibrated marginally (89.4% empirical coverage at 90% nominal) but not conditionally, under-covering in the clearest water (71.3%). Skill collapses under leave-one-zone-out extrapolation, and chlorophyll-*a*, suspended solids and thermal temperature are not retrievable. We propose two inexpensive diagnostics — a sensor classifier and null models under each candidate blocking unit — that separate genuine optical retrieval from sensor labelling and sampling design.

Keywords: Lake Titicaca, Secchi disk depth, Sentinel-2, Landsat, cross-sensor harmonization, conformal prediction, large lakes

1. Introduction

Large lakes are sentinels of environmental change, but continuous monitoring of remote, high-altitude systems is constrained by sparse field infrastructure (Williamson et al., 2009). Lake Titicaca, shared by Peru and Bolivia at 3810 m a.s.l., is the largest lake in South America and, by volume, the largest tropical lake in the world. It is a Ramsar site and the freshwater reserve for millions of Andean people, yet its water quality is documented only by infrequent field campaigns (OEFA, 2016; Siguyro Mamani et al., 2022), even as eutrophication advances in its near-shore bays (Heredia et al., 2022; Lanza et al., 2024).

Satellite remote sensing offers synoptic, repeatable observation of large lakes (Gholizadeh et al., 2016; Topp et al., 2020). Water transparency, quantified by the Secchi disk depth (Z_{SD}) since Tyler (1968), is an integrative indicator of water clarity strongly linked to the diffuse attenuation of light (Lee et al., 2015), and is among the most reliably retrievable optical water-

quality variables; it has been mapped operationally from Landsat for two decades (Kloiber et al., 2002; Olmanson et al., 2008; Zhang et al., 2022). Sentinel-2 MSI (10–60 m) and Landsat 8/9 OLI (30 m) provide complementary revisit, and their joint use as a virtual constellation is now standard practice (Claverie et al., 2018; Pahlevan et al., 2019). That joint use rests on cross-sensor harmonization, for which the per-band linear coefficients of Roy et al. (2016) are the most widely reused transform. Machine learning applied to such harmonized archives dominates water-quality retrieval across many lakes (Pahlevan et al., 2020; Cao et al., 2020, 2022; Arias-Rodriguez et al., 2023; Pahlevan et al., 2022; Hafeez et al., 2019; Tian et al., 2023).

This literature is overwhelmingly built on turbid, productive “Case-2” waters, where chlorophyll and mineral turbidity carry a strong optical signal (Gitelson et al., 2009; Mishra and Mishra, 2012; Nechad et al., 2010; Dogliotti et al., 2015). Clear oligotrophic high-altitude lakes are comparatively neglected, and two assumptions imported from that literature have not been tested in them.

The first is that harmonization coefficients derived over land transfer to water. The Roy et al. (2016) transforms were fitted on terrestrial targets — vegetation, soil and built surfaces —

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whose reflectance is one to two orders of magnitude brighter than clear water. Their additive intercepts are therefore of the same order as, or larger than, the entire over-water signal. That the transform degrades over water is not a new suspicion: a positive near-infrared bias over dark targets has been noted (Pahlevan et al., 2019), and the HLS product documentation itself cautions that its atmospheric correction is not optimized for water and that water reflectance is unreliable (Masek et al., 2021; Claverie et al., 2018). What is missing is quantification. To our knowledge no study has measured how large the mismatch is for a clear oligotrophic lake, shown what it does to a machine-learning retrieval trained on the pooled archive, or reported a routine check that would catch it.

The second is that cross-validation blocked by monitoring station is sufficient to demonstrate generalization. Accuracy in this field is routinely reported under random train/test splits that ignore spatial structure, a practice known to inflate apparent skill when samples are spatially autocorrelated (Roberts et al., 2017; Ploton et al., 2020; Meyer et al., 2019). Blocking by station is the usual remedy. It is the right remedy only if station identity is where the dependence lives — an assumption that is rarely tested and that, we show, fails in a campaign-based monitoring programme.

For Lake Titicaca specifically, prior remote-sensing work has used MODIS/Landsat temperature and chlorophyll (Ruiz-Verdú et al., 2016), basin-scale land-cover drivers (Baltodano et al., 2022), automatic Sentinel-2 water products (Ayala Izurieta et al., 2023), and global remote-sensing water-quality products (Maligaya et al., 2024). None provides a harmonized multi-mission Z_{SD} retrieval validated against in-situ measurements at their true sampling coordinates, and none assesses which variables are actually retrievable in its clear water.

We address these gaps with 1002 real satellite–field match-ups. Our contributions are threefold. First, we quantify the failure of land-derived cross-sensor harmonization over clear oligotrophic water, derive water-specific replacement coefficients from same-day paired observations, and adapt an established statistical test — the classifier two-sample test (Friedman, 2003; Lopez-Paz and Oquab, 2017), known in applied machine learning as adversarial validation — into a one-line harmonization check that any study can run. Second, we show that the blocking unit for validation must be diagnosed rather than assumed, and give the null-model procedure for doing so. Third, we report retrieval accuracy, calibrated per-prediction uncertainty and an explicit retrievability envelope under the resulting design. Our aim is not to claim a high-accuracy product but to establish what a defensible one looks like for this class of lake.

2. Materials and methods

2.1. Study area

Lake Titicaca comprises three limnologically distinct zones (Fig. 1): the large, deep, oligotrophic Lago Mayor; the shallower Lago Menor (Wiñaymarca) to the south-east; and Bahía de Puno, which receives municipal and agricultural inputs from

the city of Puno and is the most eutrophic sector. Measured transparency spans 1.5 m in near-shore turbid water to 16.5 m in the open oligotrophic basin.

2.2. In-situ data

We used 881 field measurements of Secchi transparency, with co-measured chlorophyll-*a*, suspended solids, temperature, pH, dissolved oxygen and nutrients, from the limnological monitoring campaigns of IMARPE (Instituto del Mar del Perú) on Lake Titicaca, spanning 2011–2024 across 156 stations distributed over the whole lake. Of these, 734 include a valid Secchi reading. The July 2019 campaign (71 stations) is published by Siguayro Mamani and Franco (2022); the remaining years derive from the same IMARPE programme. As a provenance check, our 2019 zone-level means reproduce the published values almost exactly (Lago Mayor conductivity 1513.4 vs. 1513.5 $\mu\text{S cm}^{-1}$; Lago Menor chlorophyll-*a* 1.52 vs. 1.52 mg m^{-3}).

Three properties of this record constrain everything that follows, and we state them here rather than in the limitations (Fig. 2).

First, the monitoring programme is itself almost entirely a dry-season programme. The 2011–2024 record contains 13 campaigns with Secchi readings, of which exactly one — December 2012, 72 measurements — falls in the austral wet season (December–March). That campaign predates both sensors used here: Landsat 8 became operational in May 2013 and Sentinel-2A in June 2015. Consequently every match-up in our dataset falls between July and October, and this is a property of the field programme rather than of our processing: no choice of match-up window or cloud threshold can recover wet-season pairs that have no contemporaneous imagery. We return to what this means for operational use in Section 4.3.

Second, sampling is episodic: campaigns exist for 2013–2024 with no campaigns in 2020, 2021, 2023.

Third, the reference measurement is coarse: 73 distinct Secchi values occur in 812 match-ups, because disk readings are recorded to the nearest half metre.

No synthetic or spectrally derived labels were used at any stage.

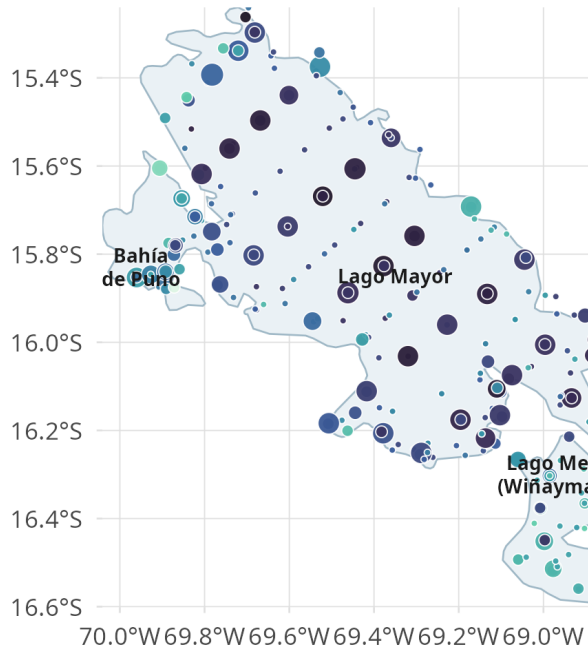
2.3. Satellite data and cross-sensor harmonization

Sentinel-2 MSI (harmonized surface reflectance) and Landsat 8/9 OLI (Collection 2, Level-2; Vermote et al., 2016) were accessed through Google Earth Engine (Gorelick et al., 2017). Clouds were masked with the Sentinel-2 Scene Classification Layer plus cloud probability, and with the Landsat QA_PIXEL bitmask (dilated cloud, cloud and shadow). The cirrus flag was not applied, as it is near-permanently set at 3810 m and would mask the entire archive. We retained the six bands common to both sensors (blue, green, red, NIR, SWIR1, SWIR2).

Landsat surface reflectance was first harmonized to Sentinel-2 equivalent reflectance with the per-band linear coefficients of Roy et al. (2016), of the form $\rho_{\text{MSI}} = a + b\rho_{\text{OLI}}$. Section 3.1 evaluates whether this transform is valid over the water of Lake Titicaca and derives an alternative. For that evaluation we identified every in-situ event observed by both sensors within the

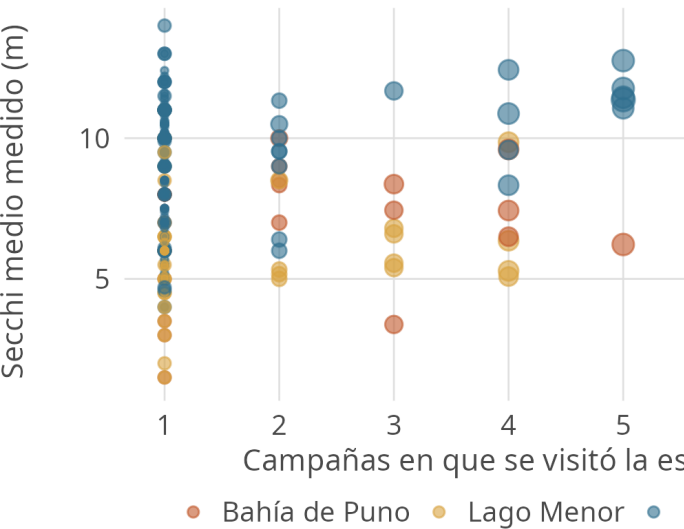
(a) Red de monitoreo y gradiente de transparencia

Estaciones IMARPE/ALT con match-up satélite-campo (2013–2024), coloreadas por transparencia media medida. El gradiente trófico e



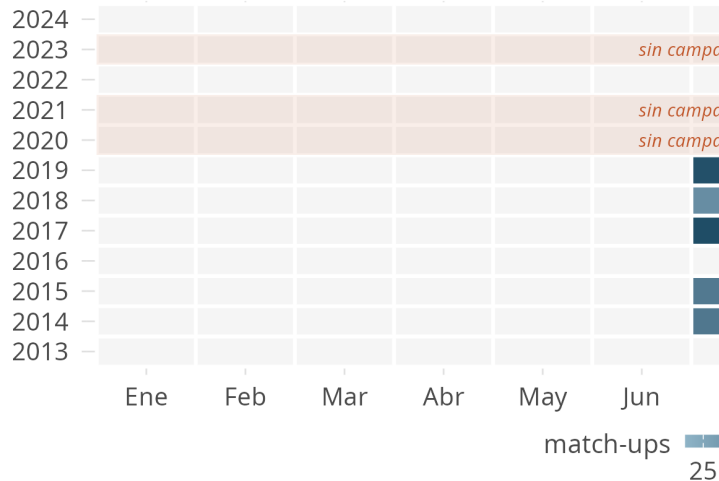
(b) El esfuerzo de muestreo es muy desigual

Cada punto es una estación. 319 estaciones, pero en muy pocas campañas, lo que limita el poder de



(a) Tres años sin campaña y ni un solo dato fuera

Número de match-ups satélite-campo por año y mes. No existió en 2013–2021, así que el hold-out «2022–2024» sólo contiene 2022 y 2024



(b) Sólo hay datos de época seca

Las floraciones de Bahía de Puno ocurren en época de lluvias, que el dataset no cubre en absoluto

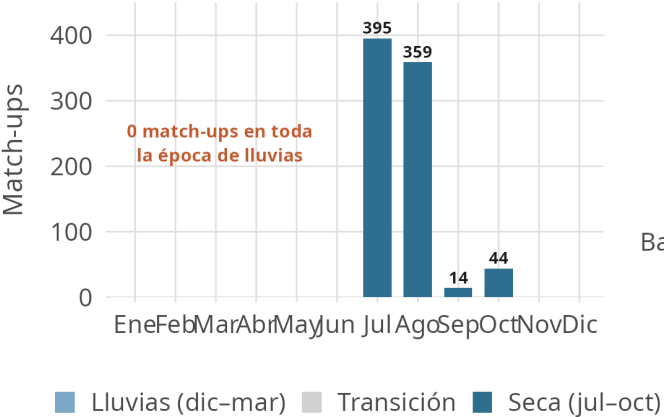


Figure 2: What the dataset actually contains. (a) Match-ups by year and month; Bahía de Puno has no campaign at all, so a hold-out described as “2022–2024” contains only 2022 and 2024. (b) Seasonal coverage: zero match-ups in the wet season, when blooms occur in Bahía de Puno. (c) Secchi distribution by trophic zone; the reference measurement is quantized at roughly half a metre.

Figure 1: Study area and monitoring network. (a) IMARPE/ALT stations with a satellite–field match-up, coloured by mean measured Secchi depth over the Natural Earth lake boundary; the trophic gradient is evident in the field data alone. (b) Sampling effort per station: most stations were visited in few campaigns, which limits the power of any station-level analysis.

match-up window — 270 paired station–date observations — and regressed Sentinel-2 on harmonized Landsat reflectance band by band.

2.4. Satellite–field match-ups

For every in-situ measurement we extracted the cloud-masked median reflectance within a 30 m buffer of the station coordinate, from imagery acquired within ± 10 days. The 30 m buffer matches the coarser Landsat pixel and averages residual geolocation error and sensor noise over several pixels while remaining in open water, away from shoreline mixing. The ± 10 -day window balances the scarce cloud-free image availability at 3810 m against the slow temporal change of transparency in this large, deep lake, and is consistent with windows used in operational Z_{SD} retrieval (Kloiber et al., 2002; Zhang et al., 2022).

This yielded 1002 match-ups (340 Sentinel-2, 662 Landsat), of which 812 have a Secchi value, across 143 stations and 120 campaign dates. Because both sensors can observe the same field event, 272 match-ups are cross-sensor pairs of a single in-situ measurement; collapsing them leaves 540 independent events, and Section 3.2 verifies that accuracy is unchanged, confirming the result is not an artefact of pseudo-replication.

2.5. Features and model

Twelve sensor-agnostic features were computed from the six common bands: the bands themselves plus NDWI, NDTI and the ratios blue/green, green/blue, red/green and blue/red. We trained a Random Forest regressor (Breiman, 2001). Reflectance was scaled with a robust (median/IQR) scaler fitted only on the training partition of each fold. All models were implemented with scikit-learn (Pedregosa et al., 2011).

Model choice was settled empirically rather than assumed. We compared Random Forest, Extremely Randomized Trees, XGBoost (Chen and Guestrin, 2016) and LightGBM, each with hyperparameters optimized by Optuna (Akiba et al., 2019) inside *nested* cross-validation — tuning on an inner campaign-blocked split of each outer training partition, so no outer fold ever informs its own hyperparameters — and benchmarked the resulting gain against the spread across random seeds. Section 3.4 reports the outcome.

2.6. Validation design and null models

Rather than a single split, we evaluate a hierarchy of increasingly demanding designs (Roberts et al., 2017): random K -fold; GroupKFold blocked by station; GroupKFold blocked by *campaign date*; GroupKFold blocked by year; a temporal hold-out; and leave-one-zone-out, in which an entire trophic zone is withheld.

Crucially, we do not assume which blocking unit matters. We benchmark each design against null models that predict, respectively, the training-set mean of the zone, of the station and of the campaign date. A blocking unit is doing real work if, and only if, the corresponding null model collapses under it. This diagnostic, rather than convention, determines our primary design.

Performance is reported as out-of-fold R^2 , RMSE and MAE, and benchmarked against a classical log-linear blue/green band-ratio algorithm in the standard form $\ln Z_{SD} \sim \ln(\rho_{\text{blue}}/\rho_{\text{green}})$ (Kloiber et al., 2002), with predictions bounded to the physically observable range, and against a multiband linear model.

2.7. Uncertainty and interpretation

We attach a per-prediction interval using split-conformal prediction (Angelopoulos and Bates, 2023): within each fold, a calibration subset of the training partition yields the $(1 - \alpha)$ quantile of absolute residuals, defining a symmetric interval with finite-sample marginal coverage guarantees (Romano et al., 2019). Because a single nominal level can be met by chance, we report empirical coverage at 50, 80, 90 and 95%, and also coverage conditional on measured transparency, since marginal calibration does not imply conditional calibration.

The model is interpreted with SHAP (Lundberg and Lee, 2017; Lundberg et al., 2020), computed *out of fold* — each observation explained by the model that did not see it — rather than on a model fitted to the whole dataset, which inflates the apparent importance of memorized features. Trends in annual median in-situ transparency were tested with Mann–Kendall and Sen’s slope (Mann, 1945; Sen, 1968) on the full 2011–2024 in-situ record, not on the match-up subset.

3. Results

3.1. Land-derived harmonization is invalid over oligotrophic water

The Roy et al. (2016) transforms carry additive intercepts of 0.0116–0.0448 in reflectance units. Over Lake Titicaca the native Landsat over-water signal is far smaller than this (Table 1, Fig. 3a). In the near-infrared the median native reflectance is effectively zero, so the intercept exceeds the signal by a factor of 69.8; in SWIR1 the factor is , in red 10.6, and even in blue — the brightest water band — 2.5. The harmonized Landsat bands are consequently dominated by a constant inherited from a land calibration: their medians exceed the corresponding Sentinel-2 medians by 1.6-fold in green and 15.4-fold in the near-infrared.

Table 1: Effect of the Roy et al. (2016) OLI→MSI transform over Lake Titicaca water. Native OLI is the Collection-2 surface reflectance recovered by inverting the transform. Reflectance is dimensionless.

Band	Roy intercept a	Native OLI median	Intercept / signal	Harmonized LS median	Harmonized S2 median
Blue	0.0183	+0.0072	2.5	0.0247	1.7
Green	0.0123	+0.0089	1.4	0.0206	1.6
Red	0.0123	+0.0013	10.6	0.0134	2.6
NIR	0.0448	−0.0006	69.8	0.0443	15.4
SWIR1	0.0306	+0.0013	23.2	0.0317	13.3
SWIR2	0.0116	+0.0013	8.2	0.0129	5.7

A harmonization that succeeded would make the two sensors statistically indistinguishable over the same water. They are not. A Random Forest classifier trained to identify the source

sensor from the harmonized features alone reaches 99.8% accuracy against a 66.3% majority-class baseline (Fig. 3b); restricting it to the visible bands gives 97.9%, and to scale-invariant band ratios 99.5%. This is a classifier two-sample test (Friedman, 2003; Lopez-Paz and Oquab, 2017): if two samples come from the same distribution, no classifier can separate them above chance. Sensor identity therefore remains fully encoded in the harmonized reflectance, and any model trained on the pooled archive can read it. The test is not new, but to our knowledge it has not been used as a harmonization acceptance check; it costs one model fit and it is decisive.

Refitting locally on the 270 station-date events observed by both sensors gives water-specific coefficients (Table 2). These restore the spectral signature almost exactly — the median Landsat/Sentinel-2 ratio falls from 15.4 to about 1.0 across all six bands (Fig. 3d) — but per-observation agreement remains uneven (Fig. 3c). Green recovers usable agreement ($r = 0.71$), and NIR and SWIR reach $r \approx 0.6$. Blue does not: $r = 0.26$. This is the most consequential result of the section, because SHAP attributes 72.6% of the retrieval to green/blue bands and ratios (Section 3.7). The band on which the retrieval most depends is the band on which the two sensors least agree.

Table 2: Water-specific LS→S2 recalibration fitted on 270 paired station-date observations, of the form $\rho_{S2} = a + b\rho_{LS,Roy}$.

Band	a	b	Pearson r	Agreement
Blue	+0.0084	0.258	0.26	unusable
Green	−0.0009	0.665	0.71	good
Red	−0.0005	0.427	0.52	weak
NIR	−0.0250	0.631	0.62	acceptable
SWIR1	−0.0223	0.770	0.64	acceptable
SWIR2	−0.0057	0.615	0.60	acceptable

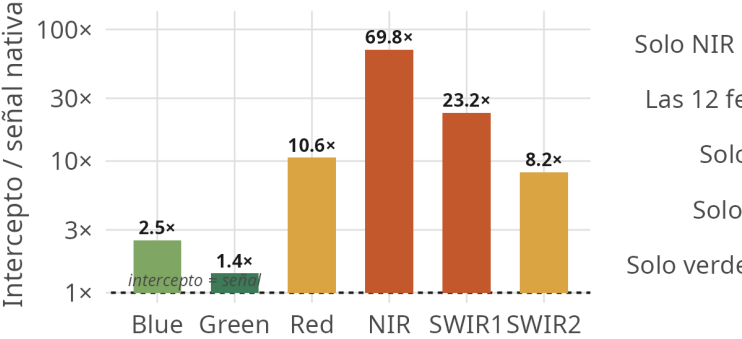
3.2. The campaign, not the station, is the blocking unit

Table 3 reports the retrieval under each validation design alongside the null models (Fig. 4). Blocking by station changes nothing relative to a random split (both $R^2 = 0.596$), which under the usual reading would be taken as evidence of no spatial leakage. The null models show why that reading is wrong: under campaign-blocked folds the station-mean null still attains $R^2 = 0.208$ and the zone-mean null $R^2 = 0.287$, whereas the campaign-date-mean null collapses to -0.012 . Dependence in this dataset lives between campaigns, not between stations, because a single campaign samples the whole lake within a few days under one set of atmospheric and limnological conditions. Station-blocked folds leave that structure intact.

Under campaign-blocked validation — our primary design — the Random Forest attains $R^2 = 0.574$ and RMSE 1.86 m over 812 match-ups, essentially unbiased (-0.02 m). Blocking by year is stricter still ($R^2 = 0.495$). Collapsing cross-sensor pseudo-replicates to 540 independent events leaves accuracy unchanged ($R^2 = 0.571$).

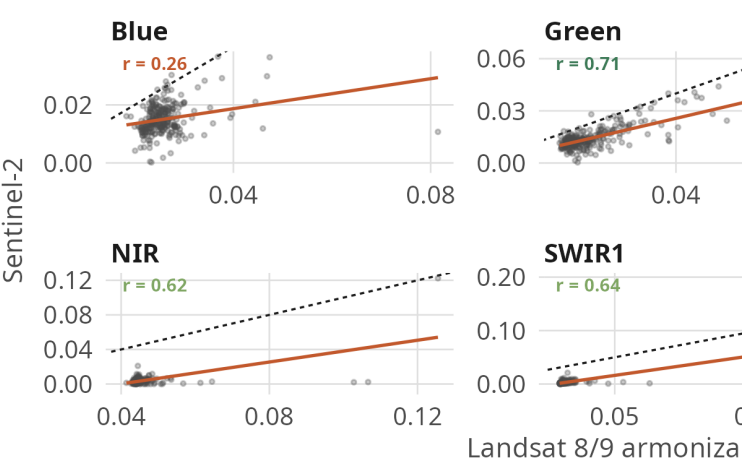
(a) El intercepto terrestre supera a la señal

Coeficientes de Roy et al. (2016), ajustados sobre vegetación y suelo, frente a la reflectancia nativa de Landsat sobre el agua del lago



(c) El azul es la banda que peor concuerda entre s

Reflectancia de Sentinel-2 frente a Landsat armonizado en la misma estación y fecha. La línea de



(d) La recalibración local sobre agua sí armoniza la

Reflectancia mediana sobre el agua por sensor. Las etiquetas pasan de 15.4x en el NIR a ~1.0x en todas las bandas

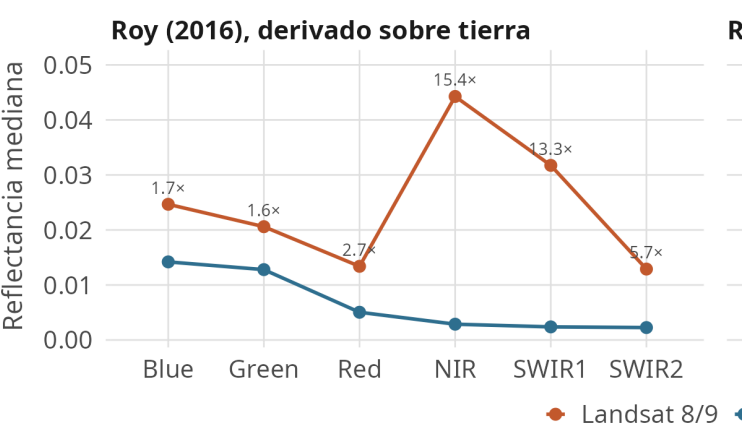


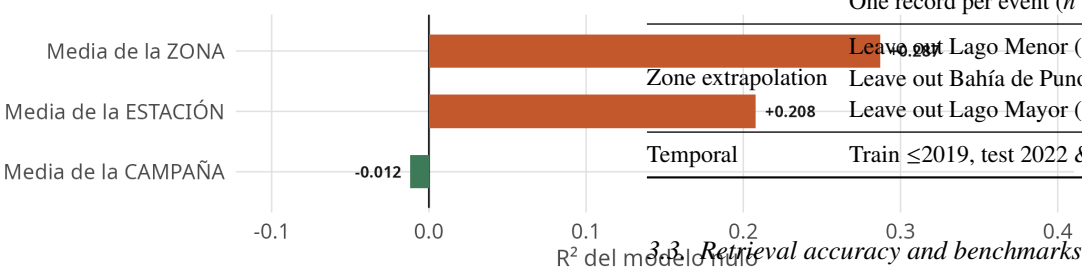
Figure 3: Why land-derived harmonization fails over clear water. (a) The Roy intercept relative to the native over-water Landsat signal, by band. (b) Accuracy of a classifier predicting the source sensor from harmonized features; under a successful harmonization it would sit at chance. (c) Sentinel-2 versus harmonized Landsat reflectance on the 270 paired station-date events; blue agrees worst. (d) Median spectral signature before and after local water recalibration; labels are the Landsat/Sentinel-2 ratio.

Table 3: Validation design, null models and blocking units (Random Forest, $n = 812$ unless noted). A blocking unit is effective only if the corresponding null model collapses under it.

	Setting	R^2	RMSE
Null models	Zone-mean	0.287	
	Station-mean	0.208	
	Campaign-date-mean	-0.012	
Blocking unit	Random K -fold	0.596	
	By station	0.596	
	By campaign date	0.574	
	By year	0.495	
	One record per event ($n = 540$)	0.571	
	Leave out Lago Menor ($n = 133$)	0.247	
Retrieval accuracy	Zone extrapolation	0.036	
	Leave out Bahía de Puno ($n = 88$)	0.036	
	Leave out Lago Mayor ($n = 591$)	-0.047	
Temporal	Train ≤ 2019 , test 2022 & 2024 ($n = 226$)	0.507	

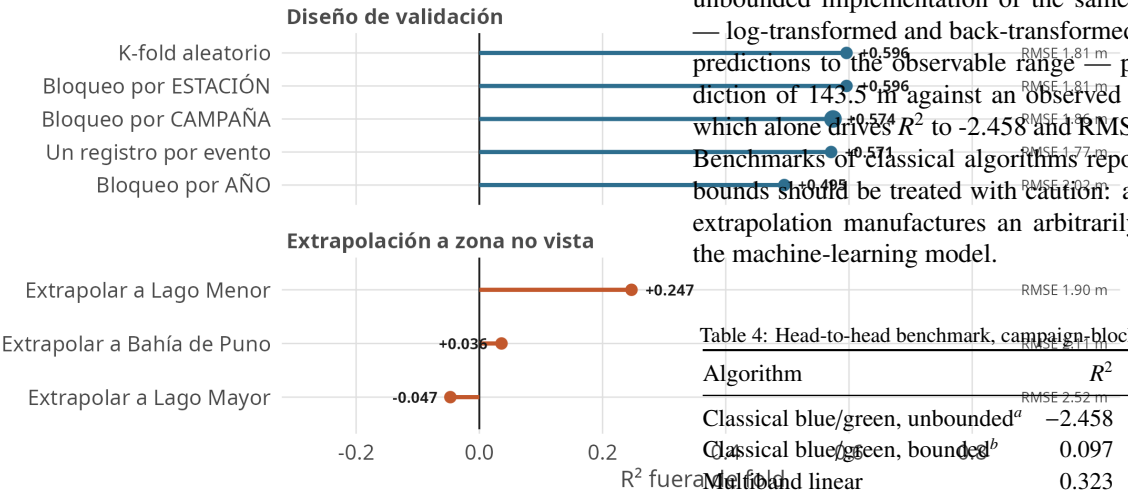
(a) Sólo el bloqueo por campaña neutraliza su propio nulo

R^2 de modelos que sólo predicen la media del grupo, con folds bloqueados por campaña. Una unidad de bloqueo sirve si, y sólo si, su propio nulo colapsa a cero bajo ella



(b) El bloqueo por estación no cambia nada; la extrapolación lo rompe todo

R^2 fuera de fold del Random Forest; el punto grande es el diseño primario. Que el bloqueo por estación coincida con el aleatorio no prueba ausencia de fuga; prueba que no retiene nada



Against this design, the classical blue/green band ratio attains $R^2 = 0.097$ (RMSE 2.71 m) and a multiband linear model $R^2 = 0.323$ (RMSE 2.34 m) (Table 4, Fig. 5). We report one methodological caution alongside these figures. An unbounded implementation of the same band-ratio algorithm — log-transformed and back-transformed without constraining predictions to the observable range — produces a single prediction of 143.5 m against an observed maximum of 16.5 m, which alone drives R^2 to -2.458 and RMSE to 5.29 m (Fig. 5d). Benchmarks of classical algorithms reported without physical bounds should be treated with caution: a single unconstrained extrapolation manufactures an arbitrarily large advantage for the machine-learning model.

Table 4: Head-to-head benchmark, campaign-blocked GroupKFold, $n = 812$.

Algorithm	R^2	RMSE	MAE	Bias
Classical blue/green, unbounded ^a	-2.458	5.29	1.91	-0.19
Classical blue/green, bounded ^b	0.097	2.71	2.13	-0.46
Multiband linear	0.323	2.34	1.78	-0.01
Random Forest (this study)	0.574	1.86	1.44	-0.02

RMSE, MAE and bias in metres.

^a back-transformed without bounding predictions; see text.

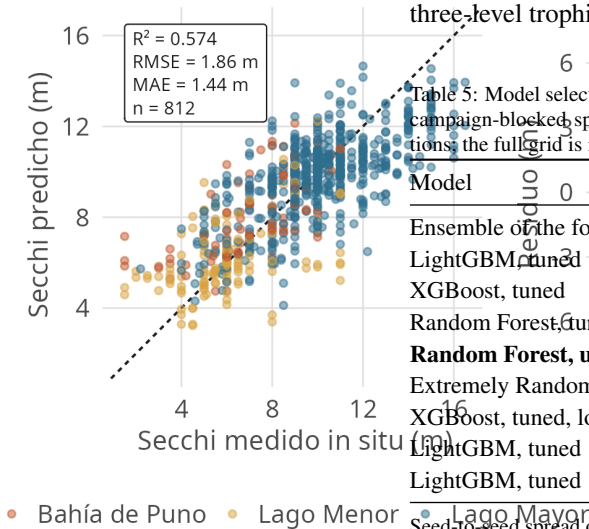
^b standard form of Kloiher et al. (2002).

Figure 4: Diagnosing the blocking unit. (a) Null models evaluated under campaign-blocked folds: only the campaign-mean null collapses, so only campaign blocking is doing work. (b) Out-of-fold R^2 under each validation design and under extrapolation to an unseen trophic zone. That station blocking matches the random split is not evidence of no leakage; it is evidence that station blocking retains nothing.

3.4. Model family and tuning do not change the picture

We compared four learner families, each tuned by Optuna inside nested cross-validation, against target transformations and reduced feature sets (Table 5). The best configuration — an equal-weight ensemble of the four tuned models — reached $R^2 = 0.583$, an improvement of 0.009 over the untuned Random Forest. That gain does exceed the spread across random seeds (0.003 in R^2), so it is not noise; but it corresponds to 0.02 m of RMSE on a lake whose reference measurement is recorded to

(a) Predicho frente a medido
Random Forest, validación bloqueada por campaña. La discontinua es 1:1



(c) El baseline estaba mal implementado
Mismo diseño de validación para todos.
El baseline «publicado» deshacía el log sin acotar; una única predicción de 143 m hundía su R^2

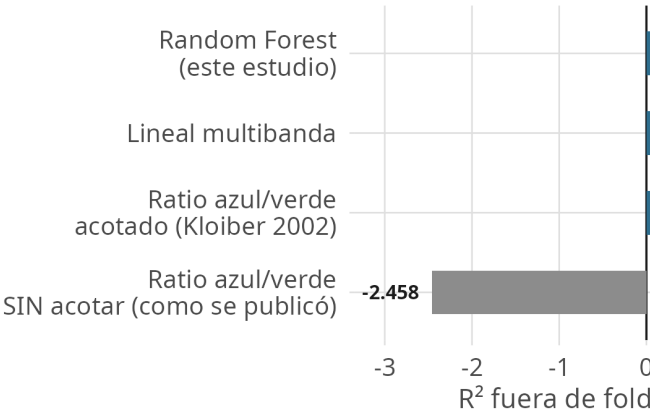


Figure 5: Retrieval under the primary design. (a) Out-of-fold predicted versus measured Secchi depth. (b) Residuals by trophic zone; the retrieval overestimates in eutrophic Bahía de Puno. (c) Benchmark against classical and linear baselines. (d) The unbounded classical implementation in log scale: one impossible prediction destroys the metric.

the nearest half metre, and it changes no conclusion in this article. Neither log-transforming the target nor tuning individual learners helped more than that, and restricting the feature set cost accuracy.

We therefore retain the untuned Random Forest as the reported model: it is simpler, faster and easier to reproduce, and reporting a tuned ensemble for a 0.009 gain would overstate the precision of the product. The ceiling here is set by the data — a dry-season-only record, a reference measurement quantized at half a metre, and a target whose variance is dominated by a three-level trophic gradient — not by the choice of learner.

(b) Sobreestima donde más importa
Residuos (predicho – medido) por zona. El rombo es la media. Bahía de Puno, la zona más eutrófica, +1.0 m

Table 5: Model selection under nested cross-validation (Optuna inside an inner campaign-blocked split of each outer training partition). Selected configurations, the full grid is in the repository.

Model	Features	R^2	RMSE (m)
Ensemble of the four tuned models	all 12	0.583	1.84
LightGBM, tuned	all 12	0.581	1.84
XGBoost, tuned	all 12	0.577	1.85
Random Forest, tuned	all 12	0.576	1.85
Random Forest, untuned (reported)	all 12	0.574	1.86
Extremely Randomized Trees, untuned	all 12	0.573	1.86
XGBoost, tuned, log Z_{sys}	all 12	0.561	1.89
LightGBM, tuned	visible only	0.544	1.92
LightGBM, tuned	ratios only	0.488	2.04

Seed-to-seed spread of the untuned Random Forest: R^2 range 0.003 over five seeds.

(d) Una predicción imposible basta
Ratio clásico en escala logarítmica.

3.5. *Where the retrieval stops working*
Withholding $R^2 = 0.46$ trophic zone collapses the retrieval (Table 3): R^2 falls to 0.247 for Lago Menor, 0.036 for Bahía de Puno, and 0.047 for Lago Mayor. The model interpolates within optical regimes, it has seen and does not extrapolate to a water type absent from training. On the temporal hold-out (train 2019, test 2022 and 2024) it retains $R^2 = 0.507$, though 57 of the 69 test stations also appear in training (82.6%), so this measures temporal, rather than fully independent generalization. Error is unevenly distributed (Table 6). The retrieval carries a positive bias of 1.01 m in Bahía de Puno, the most eutrophic and most vulnerable zone, because the clear-water majority pulls predictions upward.

Table 6: Out-of-fold error by trophic zone, campaign-blocked validation.

Zone	R^2	RMSE (m)	Bias (m)	n
Bahía de Puno	0.221	1.90	+1.01	88
Lago Menor	0.211	1.94	+0.08	133
Lago Mayor	0.448	1.83	−0.19	591

Two features of the harmonized archive should be read together with these numbers. Removing the two bands most corrupted by the land-derived intercepts lowers R^2 only from 0.574 to 0.541; using scale-invariant ratios alone gives 0.468; and applying our water-specific recalibration gives 0.518. Training on each sensor separately gives 0.563 (Sentinel-2, $n = 274$) and

0.549 (Landsat, $n = 538$). The pooled model’s small advantage over these variants is consistent with it exploiting sensor identity as an additional covariate rather than with genuine cross-sensor synergy.

3.6. Calibrated per-prediction uncertainty

Split-conformal intervals are well calibrated *marginally* at every nominal level tested: empirical coverage is 50.9% at 50% nominal, 79.1% at 80%, 89.4% at 90% and 94.3% at 95%, a maximum calibration error of 0.94 percentage points (Fig. 6a).

They are not calibrated *conditionally*, and this matters more for operational use than the marginal result. Coverage falls to 71.3% in the clearest water ($Z_{SD} > 12.5$ m) and 79.5% in the most turbid ($Z_{SD} < 5$ m), while exceeding the nominal level in the middle of the range (Fig. 6c). The mechanism is structural rather than incidental: split conformal derives a single global quantile of absolute residuals, so the interval has constant width (6.12 m at 90%) everywhere and cannot adapt to the local difficulty of the retrieval. It therefore buys its marginal guarantee by over-covering in the well-sampled middle of the range and under-covering at both extremes — which are precisely where management decisions are made. A locally adaptive variant such as conformalized quantile regression (Romano et al., 2019) would be required for reliable conditional coverage, and we recommend reporting coverage conditional on the predicted value as standard practice whenever conformal intervals accompany an operational product.

For a lake whose transparency spans 1.5–16.5 m, a ± 3 m interval is informative for zone-scale contrasts and multi-metre change, and is not informative for sub-metre differences.

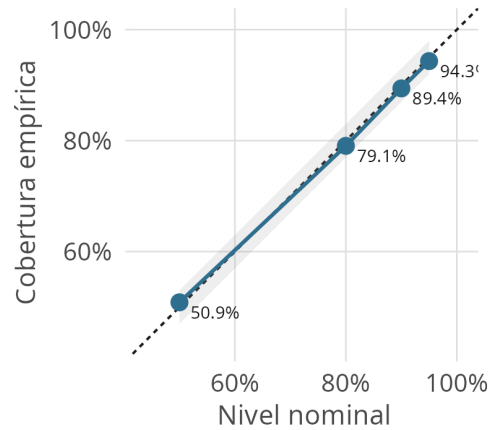
3.7. Interpretation and retrievability

Out-of-fold SHAP attributes 72.6% of the retrieval to green and blue bands and their ratios, 14.4% to NIR and SWIR, and the remainder to other indices (Fig. 7a). Green/blue alone accounts for 37%. The dependence is monotone and of the sign expected from water-clarity optics (Fig. 7b): a higher green-to-blue ratio pushes the prediction toward less transparent water. The retrieval is therefore grounded in optics rather than location memorization — while resting on the band pair with the worst cross-sensor agreement.

With real labels and the same campaign-blocked design, no other variable is retrievable (Fig. 7c): chlorophyll-*a* reaches $R^2 = 0.012$, suspended solids 0.056, in-situ water temperature 0.113, and Landsat thermal surface temperature 0.020 ($r = 0.14$, $n = 662$). Their in-situ ranges — chlorophyll-*a* 0–4.6 mg m⁻³, suspended solids mostly below 10 mg L⁻¹ — lie below the optical and thermal detection limits of these sensors. We subject the negative claim to the same design as the positive one, and report it to caution against applying chlorophyll algorithms (Gitelson et al., 2009; Mishra and Mishra, 2012) or thermal retrievals (Ruiz-Verdú et al., 2016) in clear high-altitude water without local validation.

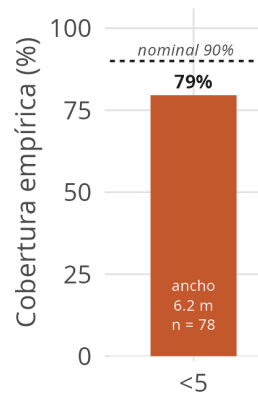
(a) Calibrados... en promedio

Cobertura empírica fuera de fold frente a la nominal. La banda gris es ± 3 puntos que la anchura es co



(c) Pero no en lo

El conformal split us no puede adaptarse



(b) El intervalo al 90% frente a la medida real

Match-ups ordenados por transparencia medida. La línea es la banda el intervalo conformal al 90%. 86 de 812 puntos que

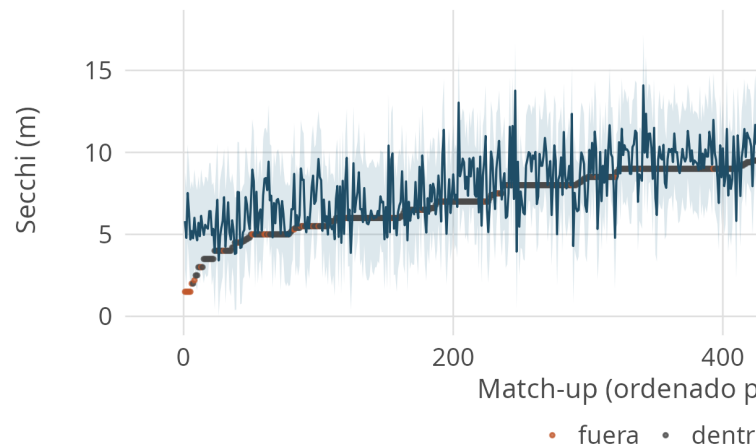


Figure 6: Conformal prediction intervals under campaign-blocked validation. (a) Empirical versus nominal coverage: marginally calibrated across the 50–95% range. (b) The 90% interval against the measured value. (c) Coverage conditional on measured transparency. Because split conformal uses a single global residual quantile, the interval width is constant (≈ 6.12 m) and cannot adapt, so the marginal guarantee is bought by over-covering in the middle of the range and under-covering at both extremes.

3.8. Temporal patterns and the fragility of trend claims

Mann–Kendall tests on annual median in-situ transparency over the full 2011–2024 record show no statistically significant trend in any zone (Bahía de Puno: Sen slope 0.16 m yr^{-1} , $p = 0.241$; Lago Mayor: 0.25 m yr^{-1} , $p = 0.138$; Fig. 8a).

That conclusion is fragile in a way worth reporting explicitly. Restricting the series to 2013 onwards — the natural choice if trends were computed on the satellite match-up subset, whose record begins with Landsat 8 — turns both Bahía de Puno ($p = 0.017$) and Lago Mayor (Sen $+0.42 \text{ m yr}^{-1}$, $p = 0.012$) significantly positive. Across start years from 2011 to 2014, 5 of 12 zone–start-year combinations change verdict (Fig. 8b).

The year 2013 has the lowest annual medians on record, so beginning there manufactures an upward trend. With 9–11 annual values and three gap years the test has low power, and the correct reading is absence of evidence, not evidence of stability. We recommend that trend analyses in campaign-based lake records report this sensitivity as a matter of course.

4. Discussion

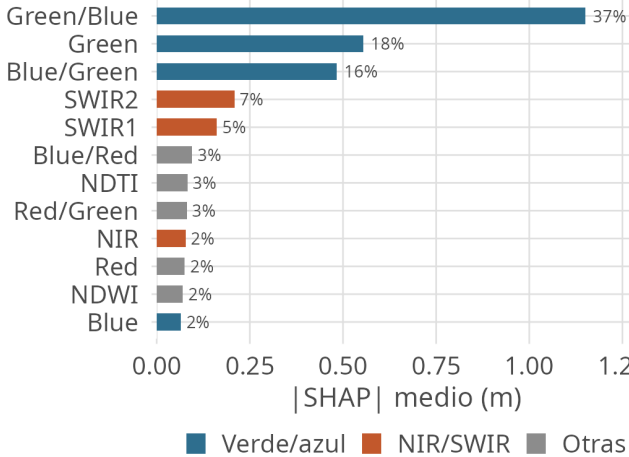
4.1. Implications for harmonized multi-mission archives

The virtual-constellation approach (Claverie et al., 2018; Pahlevan et al., 2019) has become the default for multi-decadal lake monitoring, and the Roy et al. (2016) coefficients are its most reused transform. Our results place a boundary on where that default is safe. Those coefficients were fitted on terrestrial targets and carry additive intercepts appropriate to surfaces one or two orders of magnitude brighter than clear water. Over Lake Titicaca the intercept exceeds the native over-water signal by a factor of 69.8 in the near-infrared and by 2.5 even in blue, and the two sensors remain 99.8% separable after the transform. That water reflectance from these products is unreliable is already documented (Pahlevan et al., 2019; Masek et al., 2021); what our paired observations add is the magnitude — the problem is not confined to the near-infrared, and it is large enough to make the harmonized bands function as sensor labels for a machine-learning model — and the finding that local recalibration repairs the aggregate spectral signature without repairing per-observation agreement in blue.

The practical consequence is direct. A machine-learning model trained on a pooled, land-harmonized archive can achieve apparently good accuracy while having learned two sensor-specific mappings joined by a readable sensor label. That is not a harmonized retrieval, and it will not behave as one when applied to a scene from either sensor alone. The diagnostic is cheap and we recommend it as routine: train a classifier to predict the source sensor from the harmonized features and report its accuracy against the majority-class baseline. If the sensors remain separable, the harmonization has not done its job. For clear oligotrophic lakes we further recommend deriving water-specific coefficients from same-day paired observations, reporting per-band agreement, and — where blue-band agreement is as poor as we find here — reporting single-sensor retrievals alongside any pooled product.

(a) El verde y el azul dominan la atribución

|SHAP| medio calculado FUERA DE FOLD: cada observación se atribuye a una de las características del modelo que no la vio. Las etiquetas son la cuota sobre el total de predicción hacia agua menos transparente. Record: 8 puntos extremos fuera



(c) Sólo la transparencia es recuperable en esta agua

Random Forest con el mismo diseño honesto (bloqueo por campaña) para cada variable. La afirmación negativa se somete al mismo rigor que la positiva

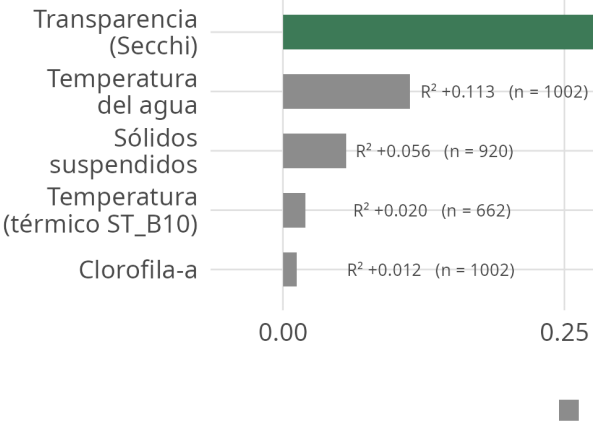


Figure 7: Interpretation and operational envelope. (a) Out-of-fold SHAP importance; green and blue dominate. (b) SHAP dependence on the leading feature, by trophic zone. (c) Retrieval of each water-quality variable under the same campaign-blocked design; only transparency clears a useful threshold.

4.2. Implications for validation design

Blocking by station is the standard remedy for spatial autocorrelation in this literature, and in our data it changes nothing: random and station-blocked folds give identical accuracy. Read conventionally, that would be reassuring. The null models show it is not. Under campaign-blocked folds the station-mean null still explains 21% of variance and the zone-mean null 29%, while the campaign-date-mean null explains none. In a campaign-based monitoring programme the whole lake is sampled within a few days under shared atmospheric and limnological conditions, so observations cluster by campaign far more strongly than by location; station-blocked folds leave that structure entirely intact.

The general point is that the blocking unit should be diagnosed, not assumed. Reporting null models under each candidate design costs little and identifies the unit that carries the dependence. This applies well beyond Lake Titicaca: most large-lake monitoring programmes sample in campaigns, so the same mismatch between the conventional blocking unit and the real dependence structure is likely to be widespread.

For our retrieval, the honest figure is $R^2 = 0.574$ (RMSE 1.86 m) under campaign-blocked validation, against 0.287 for a zone-mean null: the model roughly doubles the skill available from knowing only which basin a sample came from. That is a real but modest contribution, and stating it that way is more useful to a manager than an unqualified R^2 .

4.3. «Operational envelope for Lake Titicaca»

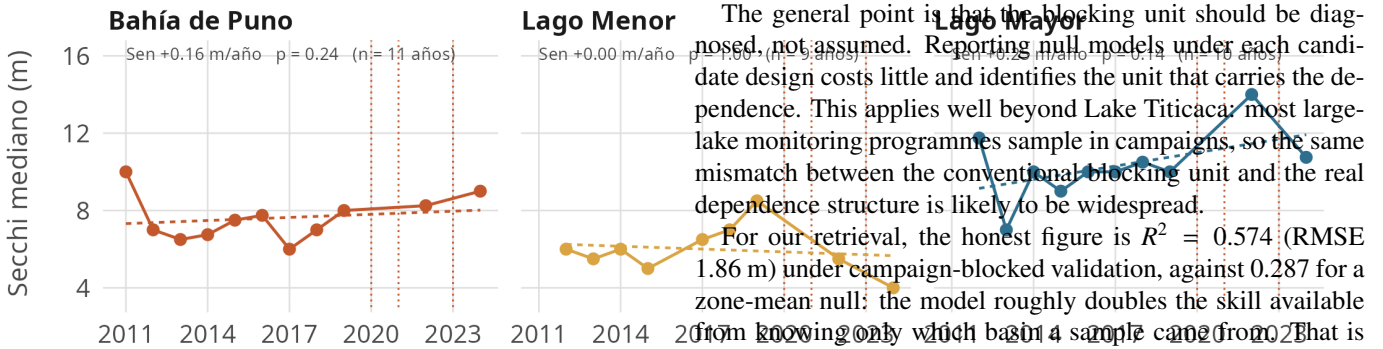
Our RMSE of 1.86 m spans roughly 12% of the observed Z_{SD} range and sits below the published range for Z_{SD} retrieval ($R^2 \approx 0.6-0.9$; Zhang et al., 2022; Lee et al., 2015), obtained here under a predominantly clear oligotrophic regime and a stricter validation design. It is also approaching the granularity of the reference measurement itself: disk readings are recorded to the nearest half metre, giving only 73 distinct values across 812 match-ups.

Four constraints bound operational use. The retrieval carries a 1.01 m positive bias in Bahía de Puno — the most eutrophic and most vulnerable zone — so mapped values there must be read with their intervals and confirmed by field sampling. Leaving one zone-out results show the model cannot extrapolate to an unsampled optical regime; integration must be incremental, each new campaign both validating the current map and extending the training set. The retrieval is calibrated and validated for the austral dry season only, and the reason is worth stating precisely because it determines the remedy. It is not that wet-season imagery is unusable at 3810 m, nor that our match-up criteria excluded it: it is that the field programme has produced a single wet-season campaign in fourteen years, and that campaign predates both satellites. Wet-season blooms in Bahía de Puno — the phenomenon a transparency product would most usefully warn about — are therefore entirely unrepresented, and the model should not be applied between December and March.

This converts the limitation into a specific, actionable recommendation for the monitoring authorities. Extending the protocol to the wet season does not require a better algorithm; it

(a) Ninguna zona muestra tendencia significativa en el registro completo

Mediana anual de transparencia in situ (2011–2024, 734 lecturas). Las líneas punteadas verticales marcan los años sin campaña: 2020, 2021 y 2023



(b) El veredicto lo fija el año en que empieza la serie, no el lago

Mismo test de Mann–Kendall, mismos datos, distinto año de inicio. Arrancar en 2013 — el año con las medianas más bajas del registro — convierte dos zonas en «significativamente positivas»

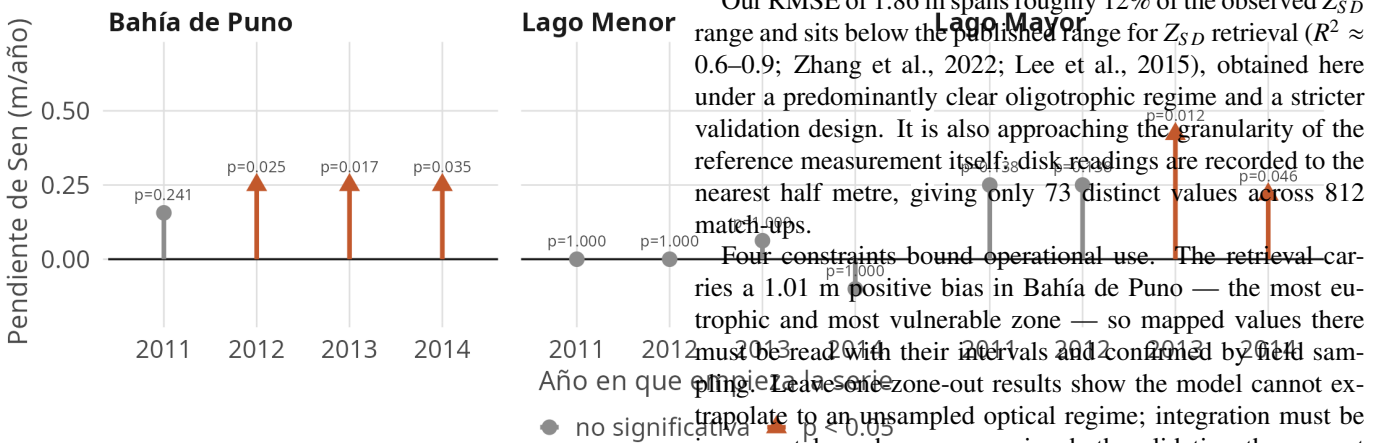


Figure 8: Transparency trends and their fragility. (a) Annual median in-situ transparency by zone over the full 2011–2024 record, with Mann–Kendall/Sen statistics; no trend is significant. (b) The same test with different start years: beginning the series in 2013, the lowest-median year on record, turns two zones significantly positive.

requires field data that do not yet exist. A single additional wet-season campaign per year, timed to any cloud-free window and covering the three trophic zones, would within a few years supply enough independent campaign-blocks to calibrate and validate a wet-season retrieval under the same design used here. Because the binding constraint is sampling rather than sensing, this is among the cheapest available improvements to Lake Titicaca's water-quality monitoring capability. And because conformal coverage degrades at both extremes of the transparency range, the nominal 90% interval should be treated as optimistic in the clearest and most turbid water.

Within those bounds the product is useful to the regional monitoring authorities (IMARPE, ALT, OEFA, and the binational management framework). A calibrated ± 3 m interval resolves zone-scale contrasts and multi-metre change, which is the scale at which clarity loss in Bahía de Puno would be actionable, and a single transboundary product under one calibration supports comparable assessment across the Peru–Bolivia boundary where coordinated field monitoring is logistically difficult. It does not resolve sub-metre change and should not be presented as doing so.

4.4. Limitations

Beyond the constraints already stated, three limitations bound this work. The in-situ record is episodic, with no campaigns in 2020, 2021, 2023 and only one wet-season campaign in fourteen years, so the temporal hold-out rests on two test years, the trend analysis on 9–11 annual values, and the seasonal envelope on none. Transparency above approximately 15 m saturates the optical signal. And this is a single lake: the two diagnostics we propose are general, but the water-specific coefficients in Table 2 are local to Titicaca and must be refitted elsewhere.

5. Conclusions

Using 1002 satellite–field match-ups over twelve years, we show that two practices imported from the turbid-water remote-sensing literature fail in a clear high-altitude large lake. Cross-sensor harmonization coefficients derived over land leave Sentinel-2 and Landsat 99.8% separable over oligotrophic water, because their additive intercepts exceed the native over-water signal by up to a factor of 69.8; and validation blocked by monitoring station leaves the dominant dependence — between campaigns — entirely intact.

Correcting both, a Random Forest retrieves Secchi depth with $R^2 = 0.574$ and RMSE 1.86 m, roughly double the skill of a zone-mean null, with conformal prediction intervals that are calibrated marginally but under-cover at the extremes of the transparency range. Skill collapses under extrapolation to an unsampled trophic zone; chlorophyll-*a*, suspended solids and thermal temperature are not retrievable at all; and no transparency trend is significant once the full in-situ record is used rather than the satellite-era subset.

We therefore recommend two routine checks for multi-mission lake water-quality products, neither of which requires

new methodology: report the accuracy of a classifier two-sample test on the harmonized features, and report null models under each candidate validation blocking unit. Both cost one model fit, and both distinguish genuine optical retrieval from sensor labelling and sampling design. For Lake Titicaca the resulting product is a dry-season, zone-scale transparency indicator with honest ± 3 m intervals — less than has sometimes been claimed for such lakes, and more than the region has had.

CRedit authorship contribution statement

Dina Maribel Yana-Yucra: Conceptualization, Methodology, Investigation, Data curation, Writing – original draft. **Richar Andre Vilca-Solorzano:** Methodology, Software, Formal analysis, Validation, Visualization, Writing – original draft, Writing – review & editing. **Fred Torres-Cruz:** Supervision, Resources, Writing – review & editing. **Vladimiro Ibañez:** Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The satellite–field match-up datasets and the complete analysis pipeline, that generate every number, table and figure in this article, are openly available.

<https://github.com/Andre031222/harmonized-titicaca-transparency>

Raw imagery is freely available via Google Earth Engine; the in-situ measurements are published IMARPE/ALT and OEFA monitoring records.



Repository

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